

Designing for Sustainable and Equitable Agricultural Practices and Technologies Between Canada and Mexico: A Multi-Phase Study with Temporary Horticultural Workers

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Abstract

Through a mixed-methods, multi-sited ethnography and participatory design approach, this dissertation examines how agricultural technologies impact temporary horticultural workers; how these workers learn, organize and transform their agricultural knowledge between Canada and Mexico, and how technology can positively intervene to support workers' agricultural practices and professional growth. This research aims to counter dominant exclusionary visions for agriculture by informing the development of epistemically and culturally aligned training for temporary horticultural workers, fostering self-efficacy, grassroots agricultural innovation, and enhanced greenhouse management and technological skills.

CCS Concepts

• **Applied computing** → **Agriculture**; • **Human-centered computing** → **Empirical studies in collaborative and social computing**; **Ethnographic studies**; • **Social and professional topics** → *Cultural characteristics*.

Keywords

digital agriculture, migrant labor, surveillance, sustainability, just transition, managed workforce, workplace technology, worker advocacy, social justice, ethnography, mixed methods

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1 Introduction

My dissertation investigates the social, political and ecological circumstances of how temporary horticultural workforces and agricultural technologies (agtechs) are designed and leveraged to uphold large-scale greenhouse practices in Canada. I argue that dominant

and historical conditions which have shaped the horticultural industry — namely scientific agriculture and racialized temporary workforces — are informing the future of greenhouses and are perpetuated in the design and implementation of agtechs. Using a mixed-methods, multi-sited ethnography and participatory design approach, I examine how agtechs impact horticultural workers; how workers in Canada and Mexico learn, organize and transform their agricultural knowledge across contexts, and if and how technology can positively intervene to support workers' practices and professional growth. I link sites between these two countries by following workers from their horticultural workplaces in Canada back to their home farms in Mexico. By following workers transnationally, I analyze local agricultural knowledge practices, social and ecological relationships, and tools. My ultimate goal is to strengthen workers' agency and autonomy, enabling them to better apply their knowledge both at home and abroad. This research aims to inform the development of culturally and epistemically aligned training for horticultural workers, fostering self-efficacy, grassroots agricultural innovation, and enhanced greenhouse management and technological skills.

2 Context and Motivation

Situated at Canada's southernmost tip, latitudinally aligned with the northern borders of California, Essex County hosts the highest concentration of high-tech greenhouses in North America [9]. This \$3 billion industry generates 503 million kilograms of produce each year [9, 10]. To maximize production yields, greenhouse owners rely on agtechs to control temperature, irrigation and other factors [10]. Owners also rely on temporary agricultural workforces. Essex hosts 10,000 workers from the Temporary Foreign Workers Program (TFWP) annually — the highest concentration of TF workers in Ontario and 6% of Canada's annual agricultural workforce [9, 28].

Since WWII, Canada's agricultural industry has endured domestic labour shortages [27]. In the 1960s and 70s, the Canadian government addressed this shortfall by developing racialized temporary workforces like the TFWP, which allowed Caribbean and Mexican people to temporarily come to Canada as farm workers with conditions well below national employment standards [27]. In Ontario, many workers do not receive adequate health care and some suffer permanent injuries and shorter careers without viable employment pathways once they return home [4, 11, 19].

Despite Canada's reliance on temporary foreign workers to maintain its food systems, these workers are often excluded from the design and use of agtechs [8, 25]. The agtechs workers do interact

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with tend to surveil and control them, and are enforced punitively to demand consistent, fast work rhythms [8, 31]. This exclusion is not just an oversight. It is tied to how agtechs are publicly defined, usually as systems that control plant and animal growth [30], leaving out human labour and its management. Although farm workers and their tools have always been critical to agriculture — from historical plantations [24] to modern agtech startups [23] — they are often left out of discussions about agricultural innovation. This omission is not a mistake, it is a deliberate design and policy choice: TF workers are viewed as a temporary fix until automation can fully take over [26]. My research directly addresses this critical omission by centering the experiences, knowledge, and needs of TF workers in the development of agtechs and related training.

3 Key Related Work & Theoretical Lenses

I situate my work within HCI and critical social science scholarship on digital agriculture and related commitments to sustainability. As shown through this corpus, digitized agricultural practices and technologies have disruptive impacts — transforming geographical landscapes of farms, labour practices, farming responsibilities and so on [3, 13]. As agtech development is often governed by the needs and infrastructures of large-scale commercial agri-businesses, small-scale farmers and agricultural workers' needs and values are often unmet [1, 22]. Part of this exclusion relates to how digital agriculture shares longstanding epistemological commitments to racialized, indentured labour and scientific research aimed at enhancing profitability and efficiency, which have shaped large-scale agricultural practices globally [18].

International agencies like the United Nations argue that agtechs should aid food security, climate crises and agri-business viability [14]. There is also an international commitment to just transitions, a concept that argues that workers displaced by sustainability development plans should have access to alternative employment pathways [16]. In transition policies, temporary workers like TF workers are not considered despite how their jobs are threatened by sustainable innovations [26].

While HCI scholars do critical work to support small-scale subsistence farmers and organic farmers, future visions for agriculture are dominated by the perspectives of Global North farmers who prioritize ecological sustainability over other sustainability missions like food security [21, 29]. My dissertation helps to bridge this gap by leveraging TF workers' transnational agricultural knowledge to encourage sociotechnical systems that balance economic, ecological, and social sustainability in ways that are inclusive to small-scale farmers across development contexts.

My research is informed by the theoretical areas of Science and Technology Studies (STS), feminist technoscience and postcolonial computing [12, 17]. I rely on these lenses to maintain that technologies are value-laden and shaped by dominant political, social and economic systems. By situating agricultural knowledge in local contexts, I show how workers hold multiple types of knowledge that are partial, evolving and shaped by power relations. By learning from TF workers' knowledge practices, I hope to inform more culturally and epistemically aligned training and professional pathways for horticultural workers.

4 Research Questions

My project addresses the following questions, drawing from fieldwork across Canada and Mexico:

- (1) How are temporary horticultural workers impacted by agricultural technologies in Ontario, Canada?
- (2) How do temporary horticultural workers describe, learn and organize their agricultural knowledge in Canada and Mexico and how does this knowledge transform between contexts?
- (3) If and how can technology serve as a positive intervention to support temporary horticultural workers in Canada and Mexico?

5 Methods

To answer the above questions, I have divided my fieldwork into three research phases: one phase of mixed methods discovery fieldwork with TF horticultural workers in Ontario, one phase of multi-sited ethnographic fieldwork with these same workers between greenhouses in Ontario and their home farms in Oaxaca and Veracruz, Mexico and one final phase of participatory design workshops with TF workers in Ontario. All research with TF horticultural workers has been conducted in Spanish by myself or a Spanish-speaking community representative. Throughout my project I hold the transformative aim to support technologies and professional pathways for temporary workers. I rely on Merten's [20] transformative research framework to situate this change agenda, which emphasizes centering impacted communities' experiences and perspectives while connecting power inequities to material sociopolitical actions.

For the first phase, I collaborated with a Canadian labour union to co-design an embedded mixed methods study composed of an exploratory survey and in-depth interviews with temporary agricultural workers. With little existing research on how agtechs impact this group, these activities were foundational for better understanding workers' interactions and attitudes toward agtechs and their desires for improved workplace technologies and practices.

For the second phase, I followed five TF workers back to their home farms in Mexico to understand how workers learn in their home contexts, how agricultural knowledge is organized and how worker knowledge transforms across transnational contexts. I conducted a multi-sited ethnography composed of site visits, observation of participants' farming activities and in-depth interviews.

For the final phase, I am conducting two participatory design workshops with TF workers in Essex County, Ontario in Summer 2025. Informed by the results from my two first phases, these workshops will be centered on understanding workers' specific learning needs in agricultural management and their ideal learning styles, providing hands-on tutorials in irrigation methods, fertilization management and climate control techniques and sharing affordable, scalable and sustainable agricultural practices that TF workers can hopefully implement in their home countries. By the time of the *CSCW 2025* conference, I will have completed all three research phases.

6 Research Findings and Progress to Date

I have published three peer-reviewed papers from my dissertation with two more under review. My first publication highlighted a

speculative design project I created, which aimed to provoke consumers to consider what food packaging would look like if labour conditions were emphasized over other details like organic standards [5]. My second article was a literature review where I found that digital agriculture in HCI often focuses on the needs of Global North organic farmers, even though most studies address the needs of Global South subsistence farming [6]. My third paper, from my first fieldwork phase, revealed that surveillance and labour tracking technologies significantly impact horticultural workers in Ontario. These agtechs cause stress due to performance expectations, fear of disciplinary action, and increased competition [8]. Workers also expressed a desire for more technical training and transparency about personal data collection. I have also co-organized three HCI panels on human labor, automation, and environmental justice [2, 7, 15].

My paper from the second research phase is currently under review, and I am planning my final design workshops for Summer 2025. Beyond publications, I am actively involved in knowledge mobilization. This includes co-organizing an international symposium on "Just Transitions" in the greenhouse industry in Mexico and working with union and government representatives to turn my research into policy recommendations and public reports.

7 Proposed Contributions and Open Questions

The CSCW 2025 doctoral consortium offers me a chance to think deeply about how I write about my final fieldwork and to reflect on my dissertation's framework and its broader impact. As a senior PhD candidate, I believe that I can contribute to the consortium by sharing my experiences of conducting community-engaged fieldwork in a second language and leveraging various research methods including an embedded mixed methods study, a multi-sited ethnography and participatory design workshops. I can contribute to conversations regarding how CSCW may support temporary or marginalized workforces, particularly those who are denied employment rights and who live between multiple international contexts.

Below I list three pressing questions from my research which I hope to discuss at the consortium:

- (1) Traditional CSCW tools (e.g. design interventions) are not always suitable to address the complex social and environmental problems our world faces. How do others think CSCW's contributions and necessary collaborations should change to address these diverse issues?
- (2) Beyond the tension of usefulness versus solutionism in material technology development, how do we ethically account for and actively support marginalized communities who are often unresourced to build the technologies they need?
- (3) In my multi-sited research, I have observed that sustainability and equity are defined very differently across Global North and Global South contexts. How can we bridge these conceptual gaps to design and work effectively across these distinct development contexts?

8 Dissertation Status and Long-Term Goals

I am a fourth year PhD candidate at the Faculty of Information and the School of Environment at the University of Toronto. By the time of the CSCW 2025 conference I expect to be writing my

final dissertation chapter. I anticipate defending my dissertation by Summer 2026. After graduating, I hope to obtain a research and teaching position at an academic institution in Canada. I am interested in information science, environmental studies, computer science and organizational management.

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